LR → y

• Polynomial Regression ⇒

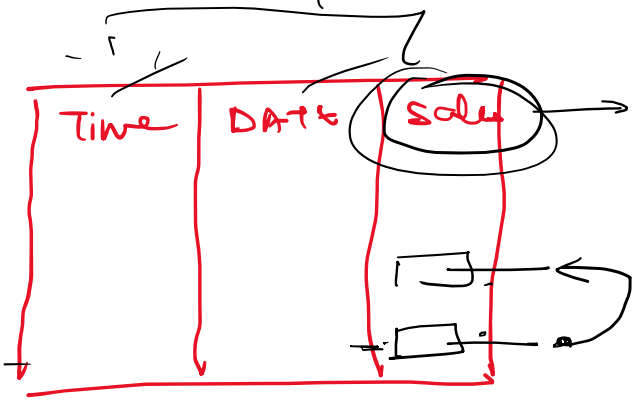
. Agenda ⇒

- → No autocorrelation →
- → Polynomial Reg ⇒ extn of LR
- → generalization
- → underfitting & overfitting

No autocorrelation ⇒

x used → time series

→ Time series



→ o/p variable

LR

→ Autocorrelation

x → i/p

o/p

→ Sales tomorrow depends on Today?

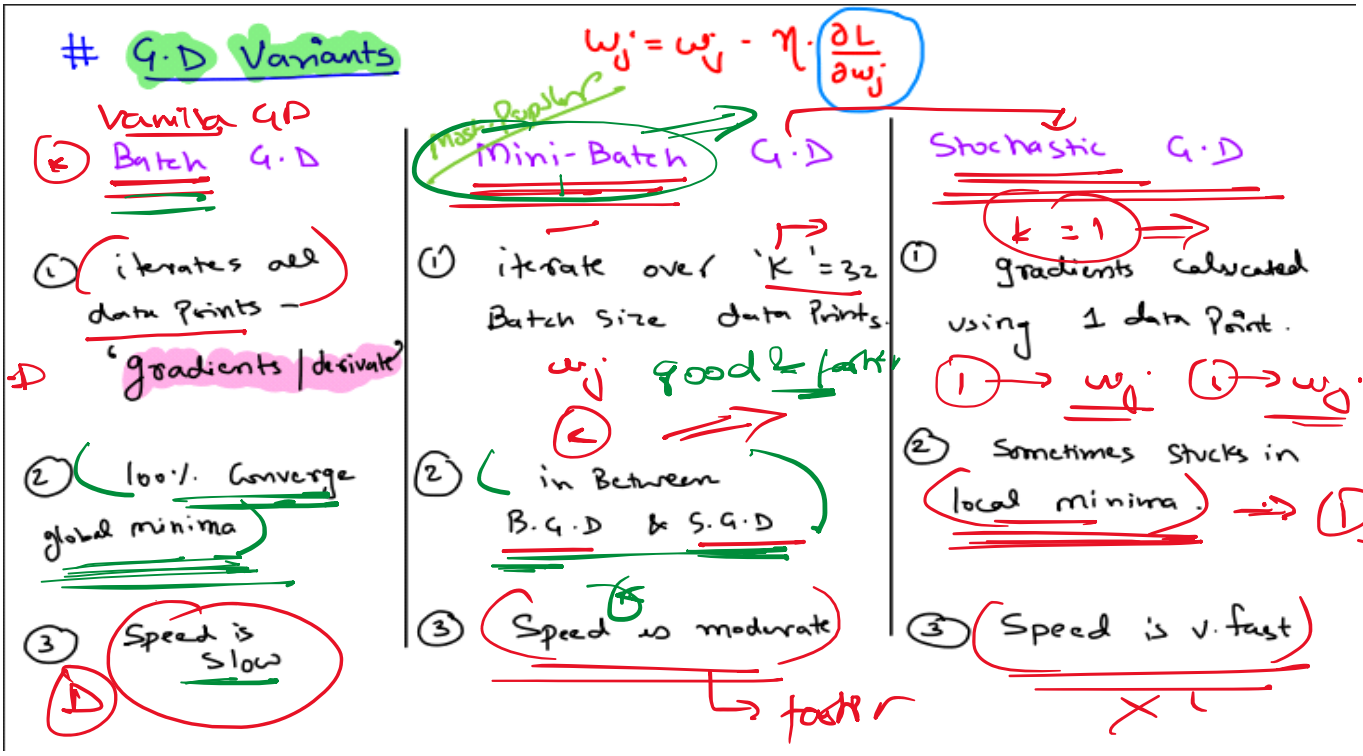
LR → there should be no dependencies of R_{β} with each other

DP → should be independent of each other

(X) Time series

GD variants ⇒

GD → $w_j = w_j - \eta \cdot \frac{\partial L}{\partial w_j}$ (4)



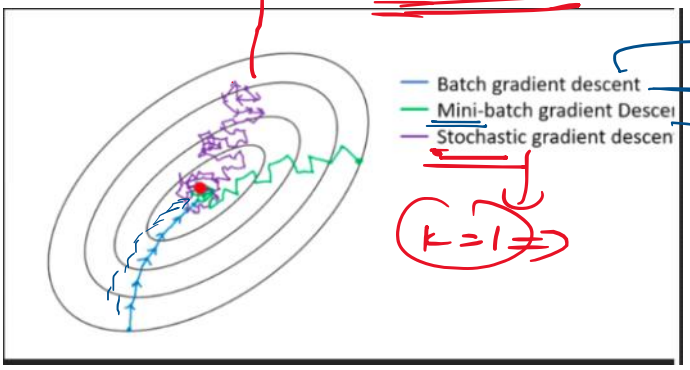
(1) → $k = 32$ → w_j

D → k = 32 → wj

32 → wj

32 → wj

Randomness ⇒ (1) BP



smooth

changing dir^n in B/w

some fluctuation

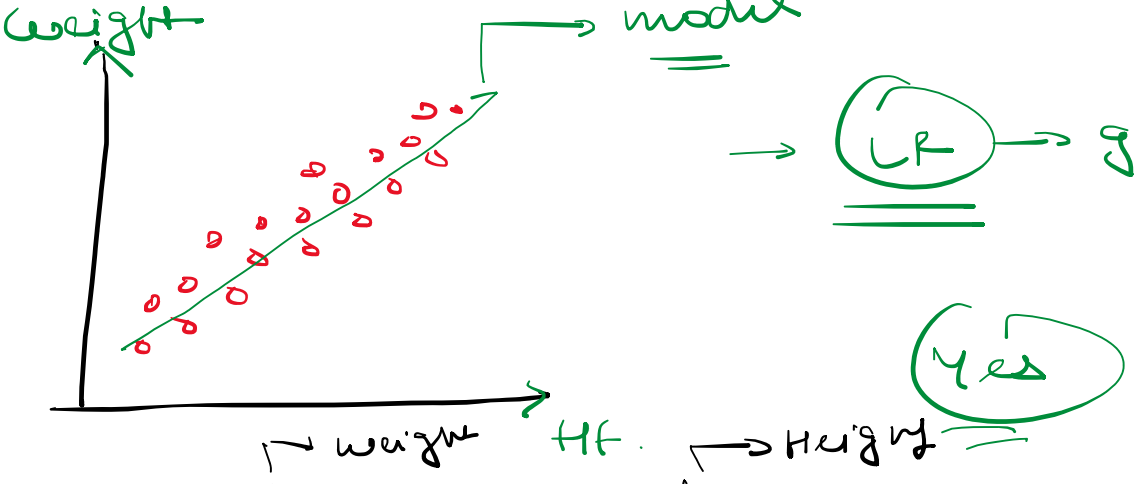
k=1 ⇒

DL ⇒ Non linear ⇒

ML →

(*) NN → layers

noisy ⇒ STOCHASTIC (*)



model

LR → good

Yes

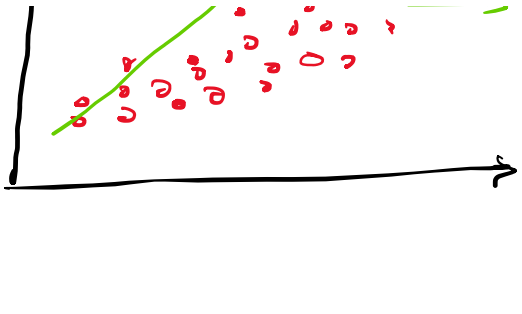
wt. ∴ $\hat{y} = w_0 + w_i f_i$

linear model

X → X

score ↓

DATA = NON linear



Model = Linear

Score ↓

what will i do?

Hack → f_1^2

f_1	f_1^2	Y

→ LR model

2 i/p

Y

2nd one → Non Linear feature

↳ connect to the o/p variable

f_1^2

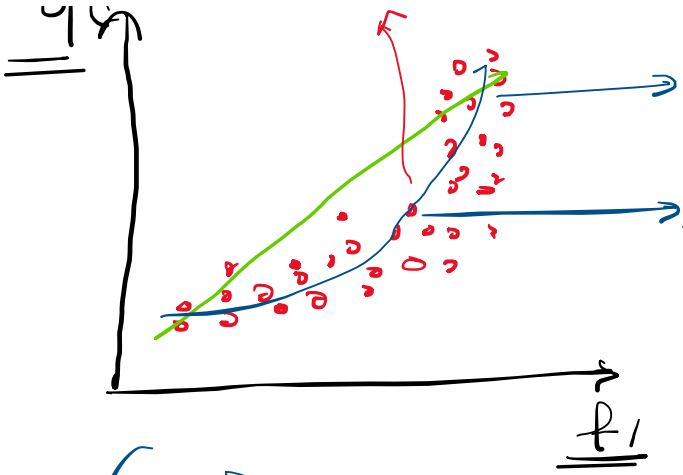
LR

o/p

curve

o/p → Better models

Complex



Better models

eq^

LR

Complex

$\hat{y} = w_0 + w_1 f_1 + w_2 f_1^2$

updated model eq^n

LR → Non Linear DATA

algo remains same ⇒ same LR

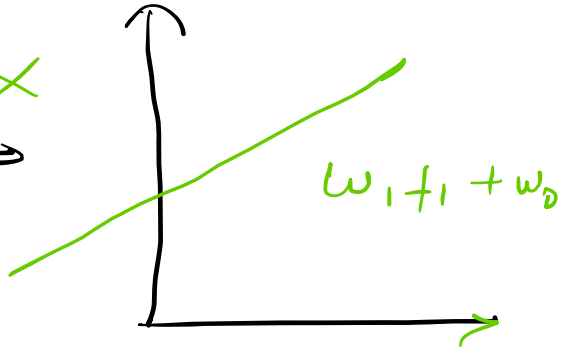
Q What Happens ?

model → Non Linear → Non Linearity → Non Linear

Non Linear ⇒ SVM, kernel

f1	y

LR ⇒ X

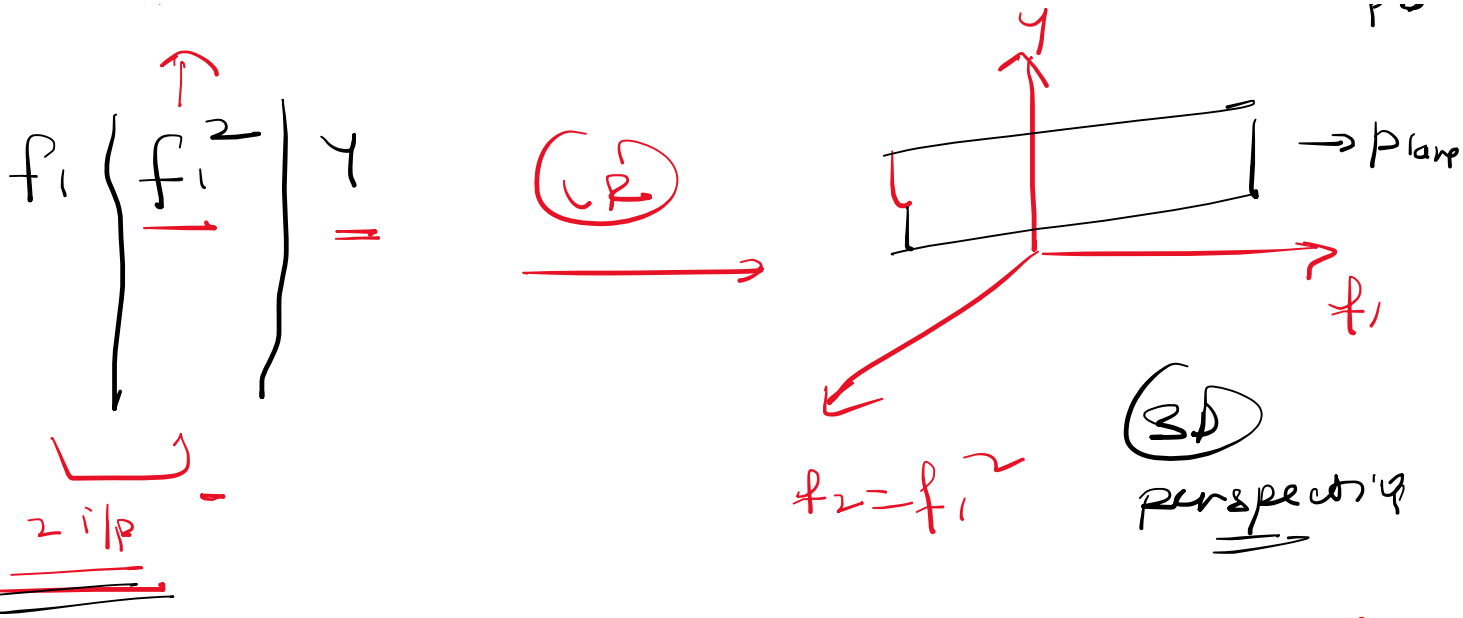


Q transformations

2D → line

3D → plane

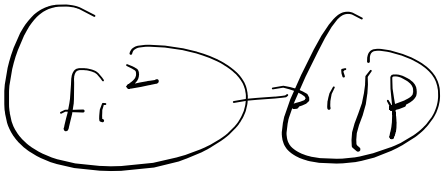
4D → hyper plane

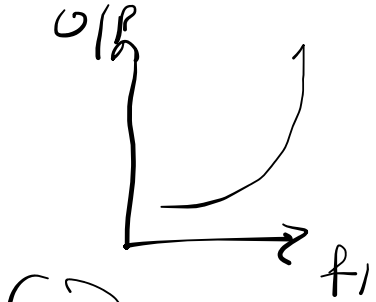


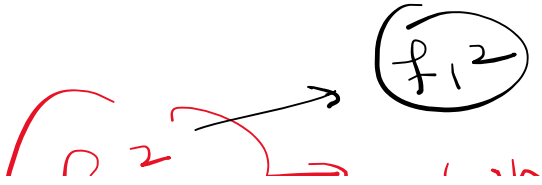
$$\hat{y} = w_1 f_1 + w_2 f_1^2 + w_0$$

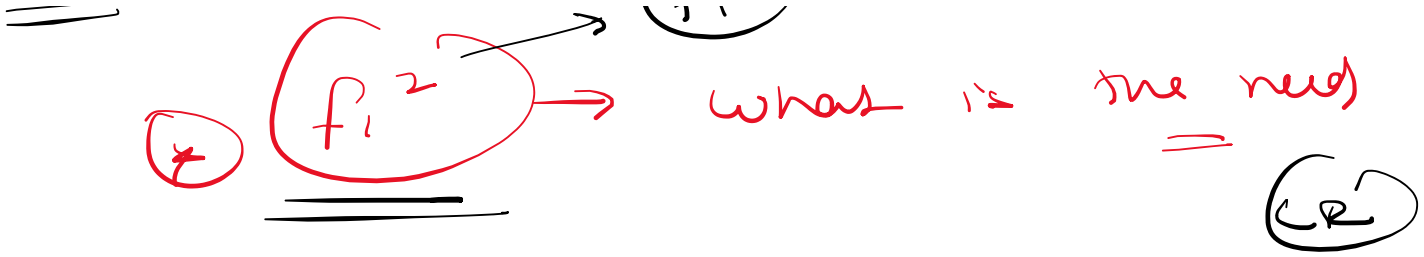
(2) model → plane now

(2)

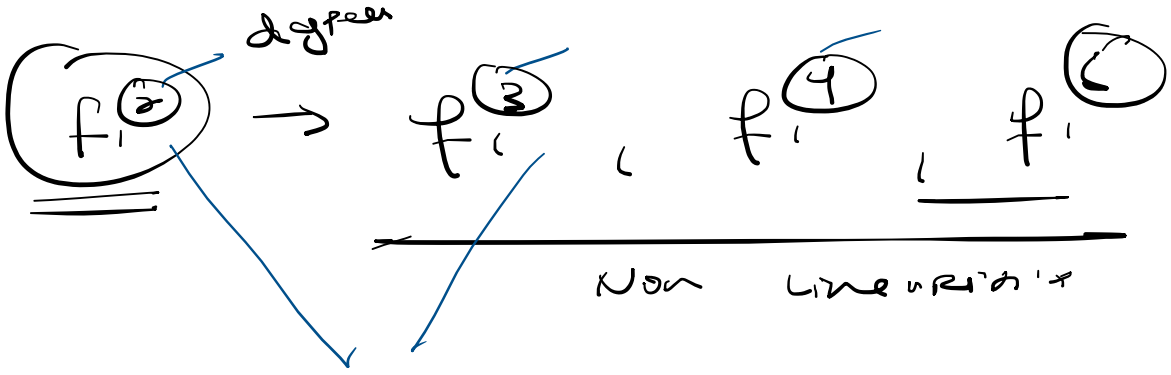








(*) Real world ⇒ NON Linear



(*) Hyperparameters → Hyperpara

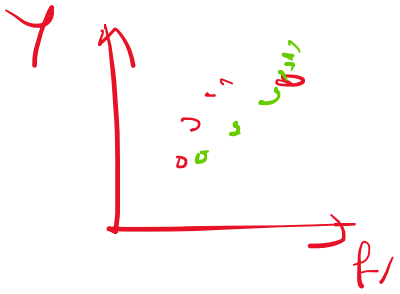
(1 - 0) degree

↳ performance

80%
Linear
2 → 99%
3 → 99%
4 → 99%

(1) y f_1 → pred

Numerical



Numerical

Age, Height

f_1^2 f_1 f_2

$f_1 \times f_2$ Non linear

f_1 f_2 f_3 f_4

$f_1 f_2$ $f_1 f_3$

$\text{Adj } R^2 \rightarrow$ ↑ ↓

$f_1 f_4$ $f_1 f_2 f_3$

Does $f_2 = f_1^2$ cause MC ?

X

MC → linear combⁿ of existing features

↳ $f_2 = \alpha(f_1^2) + \beta$ → Linear → MC

f_1^2 → Non Linear combo

④ LR

age, Year

X → MC

$age_2 = Year^2 + 2020$ (α on Year, β on 2020)

age × Height

↓ ↓

f_1 f_2

$f_1 \circ f_2$ // Non Linear Comb

⑧ $f_1^2 = f_1 \times f_1$

$f_2 = f_1 \times f_1$ → Non Linear

↓

MLR ✗

↓

Feature engineering

FE → techniques

LR ⇒

$(f_1 \quad f_1) \Rightarrow f_1^2 \quad f_1 f_2 \quad f_2^2$

↓

R^2 & Adj R^2

performance

$R^2 \uparrow$ Adj $R^2 \downarrow$

↓

✗

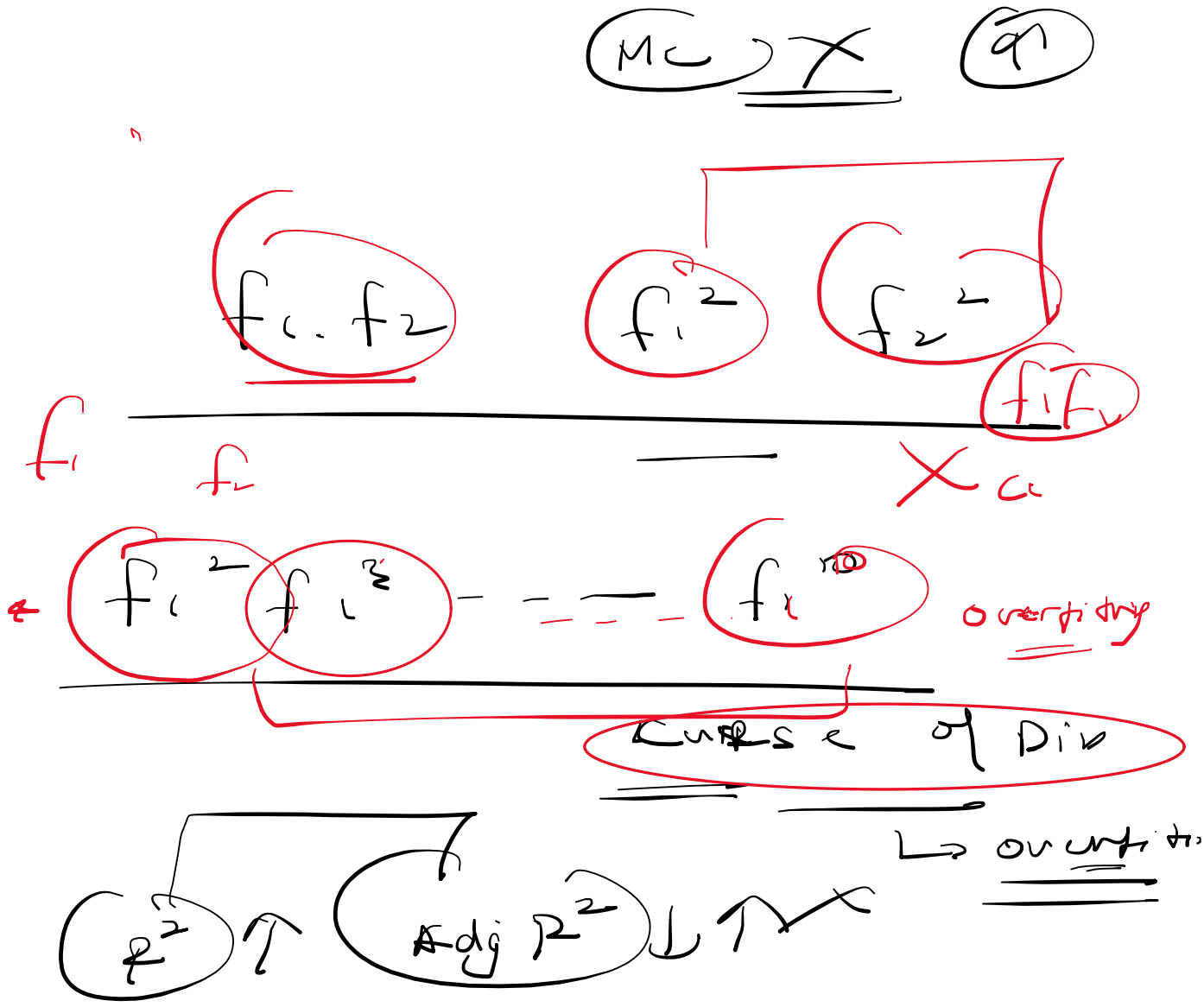
✗

EDA

FE

ML → time consuming

FE → Non Linear features



Time Left 22s

Which statement is true about mini-batch gradient descent?

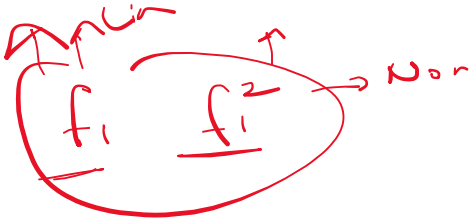
23 users have participated

A	It guarantees convergence to the global minima	43%
B	It may converge to a local minima due to the weight fluctuations	57%
C	It requires a very high learning rate.	0%
D	It is not suitable for large datasets.	0%

End Quiz Now

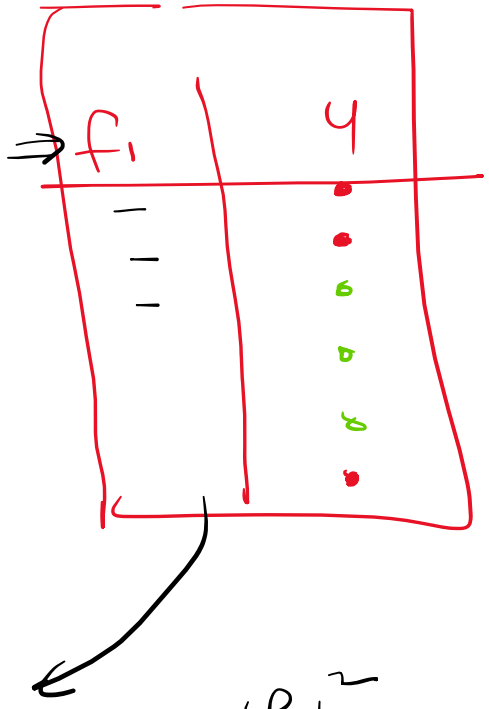
② Perceptron

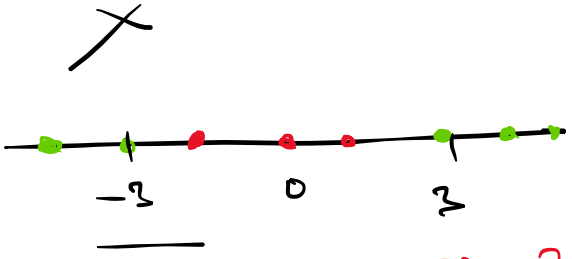
③ Classification



eqn → Linear
Non

④ Classification





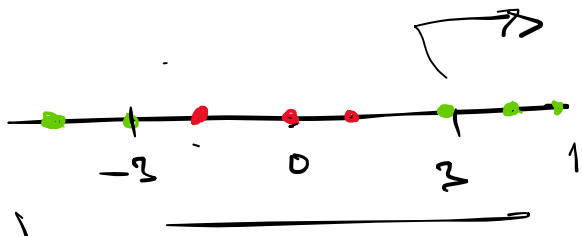
① linearly seperable ?

FE

f_1^2

f_1 → f_1^2 y → DATA looks now

X → Non linear



→ Non linear data in lower

(+) 1D

f_1 f_1^2 Y

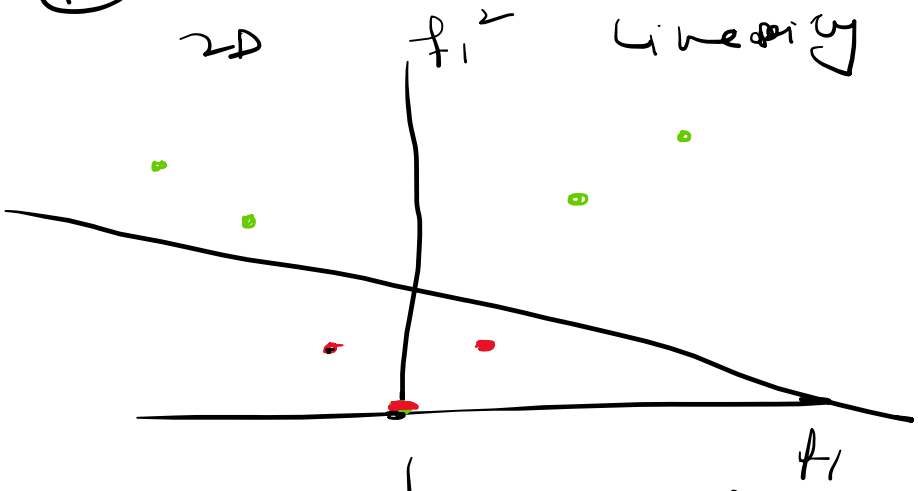
0 → 0

1 → 1

-1 → 1

3 → 9

-3 → 9



↳ seperable now

(2D) (Plane)

Learning ⇒

(+) Non linearity

2D → Parabola → curve

3D → Plane

Non Linear data in lower dim

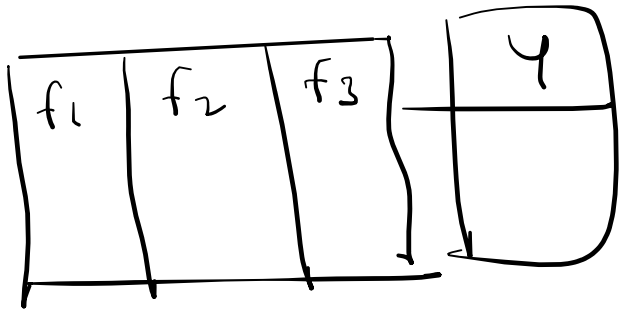
=

Linear data in higher dim

4

10:16

3D



LR

$\Rightarrow$ $R^2 = 0.65$

Poor or good

poor

what can i do?

FE → $f_1^2 + f_1^3 + f_1^4 + \cdots$

$f_1 \cdot f_3$

experiment with any

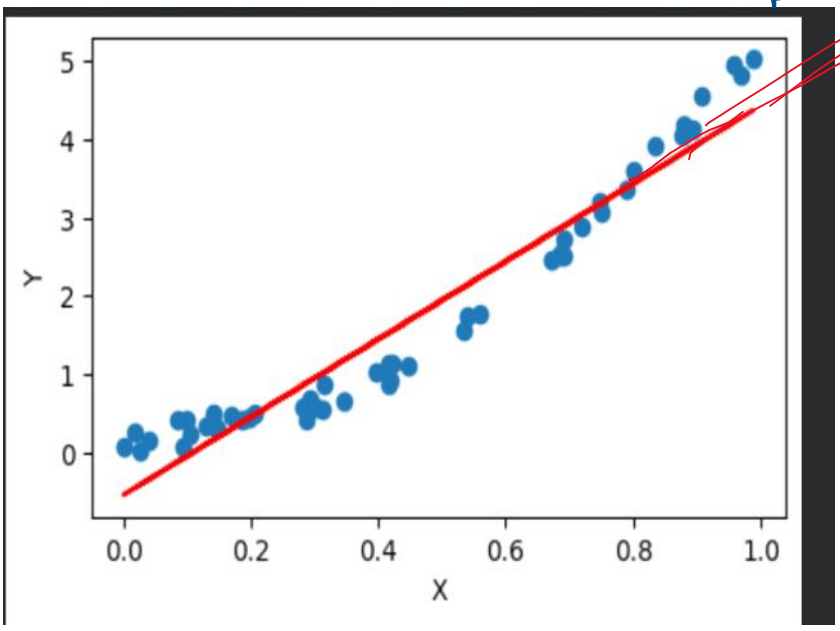
LR → $R^2 = 0.79$ → 12%

good ?

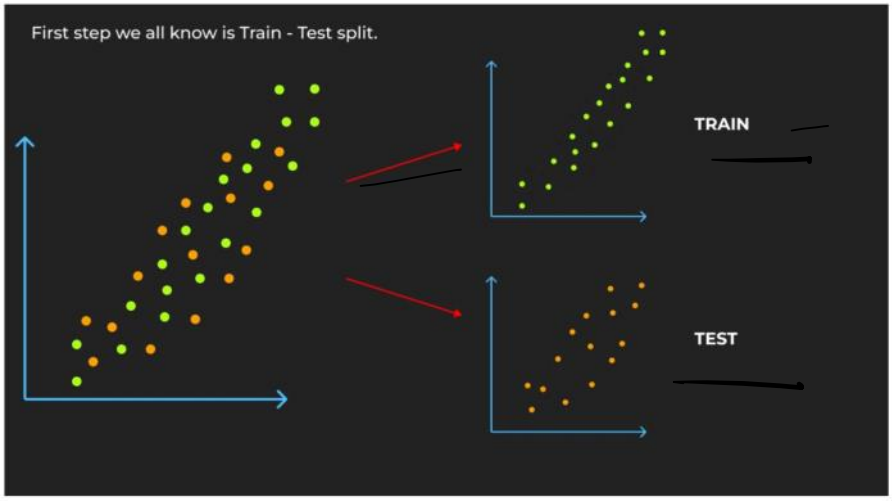
Polynomial Reg^n =

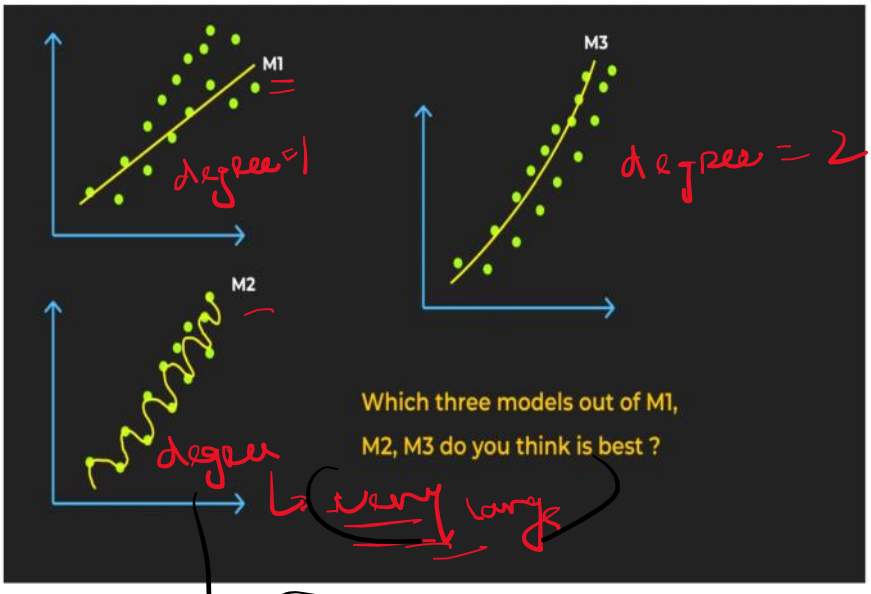
Linear Reg^n + (Non linear features Reg^n)

$\hat{y} = w_0 + w_1 f_1 + w_2 f_2$


5
4
3
2
1
0
Y
0.0
0.2
0.4
0.6
0.8
1.0
X

Generalization ⇒





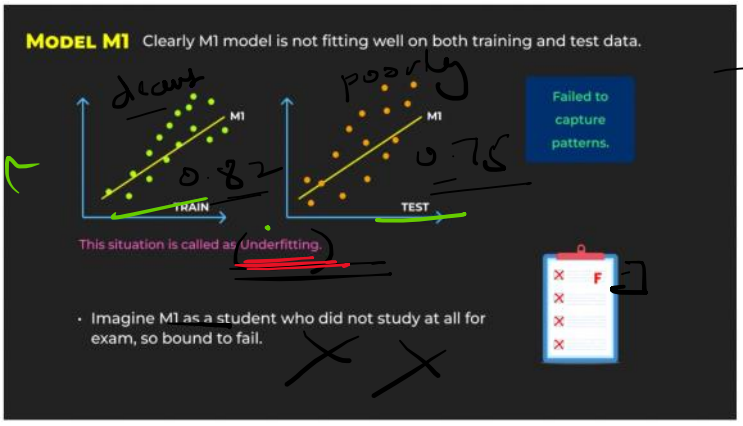
Train

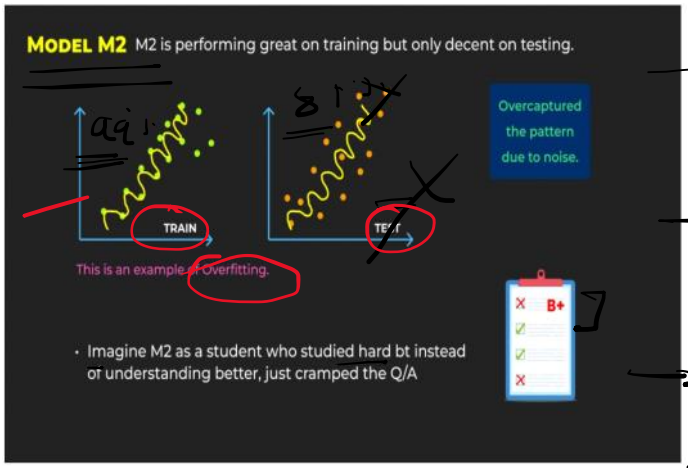
Train

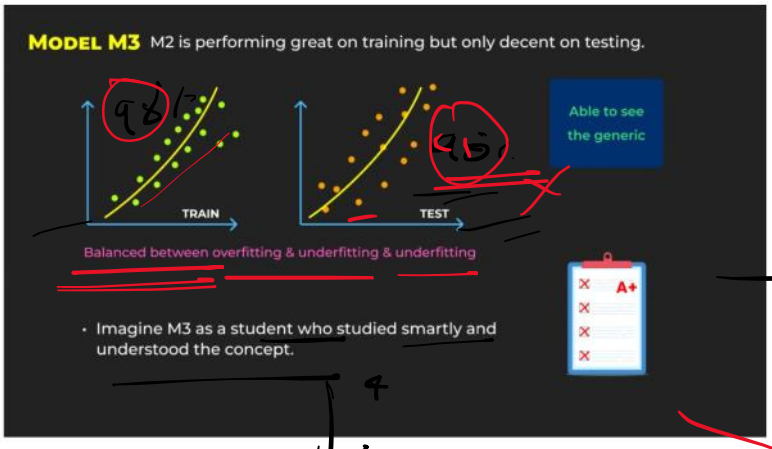
Train

M2 → ↑

unseen data


MODEL M1 Clearly M1 model is not fitting well on both training and test data.
M1
TRAIN
M1
TEST
Failed to capture patterns.
This situation is called as Underfitting.
Imagine M1 as a student who did not study at all for exam, so bound to fail.
F


MODEL M2 M2 is performing great on training but only decent on testing.
TRAIN
TEST
Overcaptured the pattern due to noise.
This is an example of Overfitting.
Imagine M2 as a student who studied hard bt instead of understanding better, just cramped the Q/A
B+


MODEL M3 M2 is performing great on training but only decent on testing.
TRAIN
TEST
Able to see the generic
Balanced between overfitting & underfitting & underfitting
Imagine M3 as a student who studied smartly and understood the concept.
A+

Healthcare → 98% → 11 PM

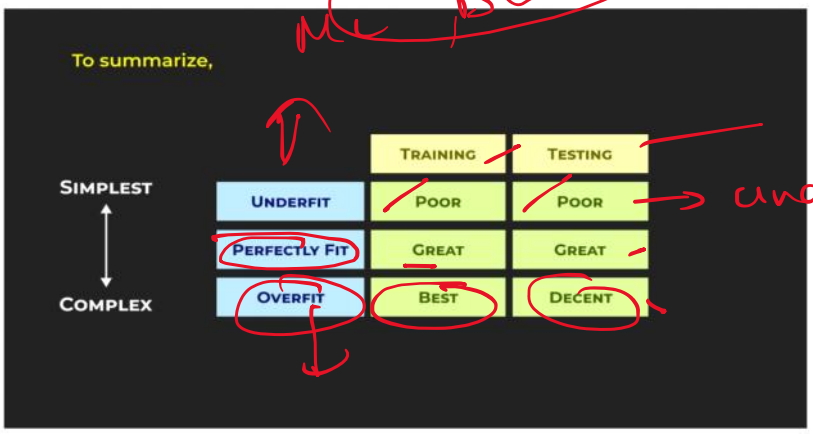
→ Spend

Fraud → 80% → ↑

Payment

Gu

ML DL AI



underfit

degree = 10

$w_0 + w_1 f_1 + w_2 f_1^2 + \cdots + w_{10} f_1^{10}$

greedy

Overfitting

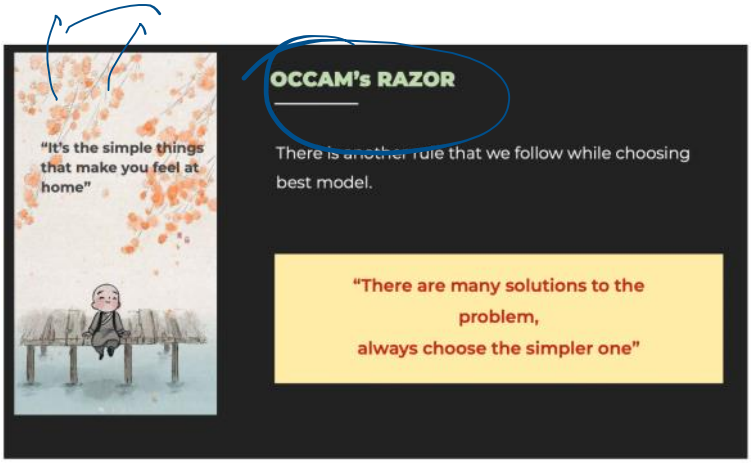
Generalization ⇒ Test data performance matters also

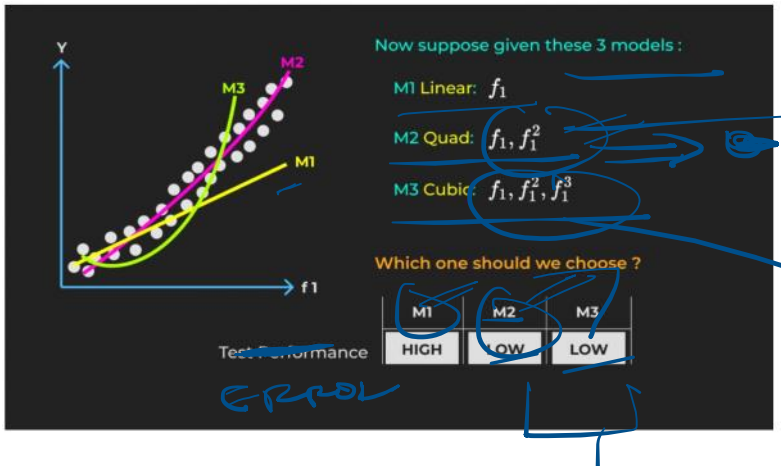
unseen data

OKAMS RAZOR ⇒

problem → multiple solns

Simplest to implement





w_1 w_2 → Simpler

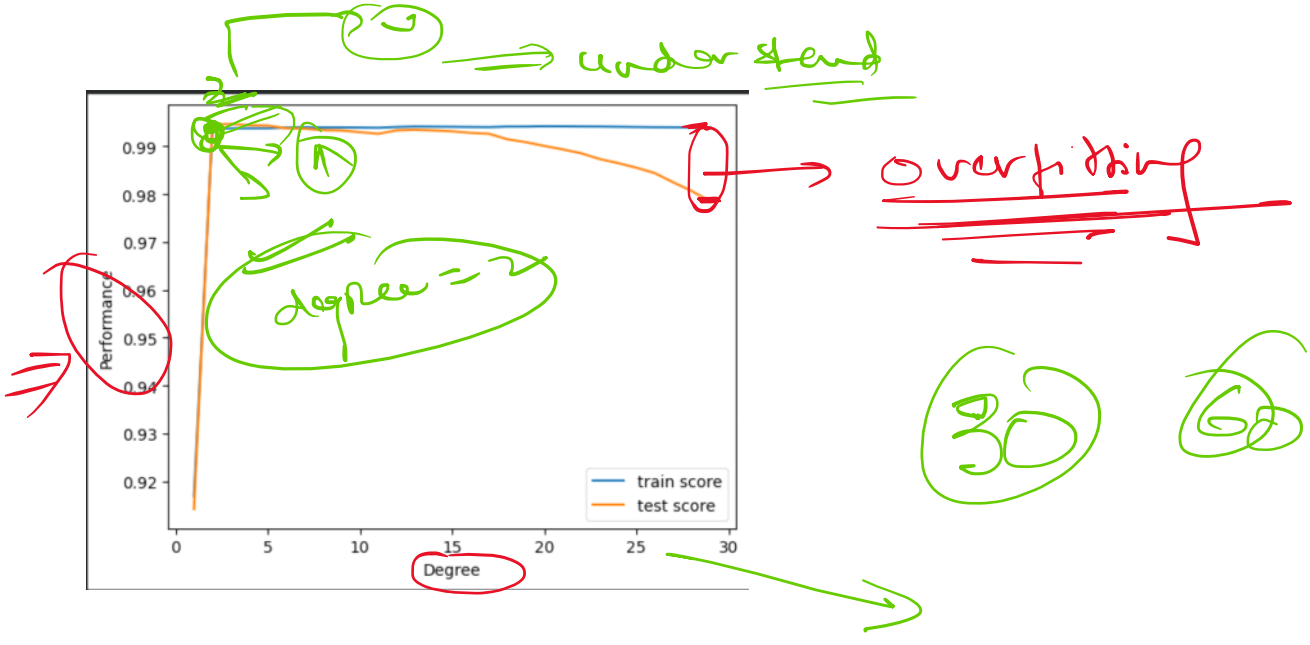
w_1 w_2 w_3

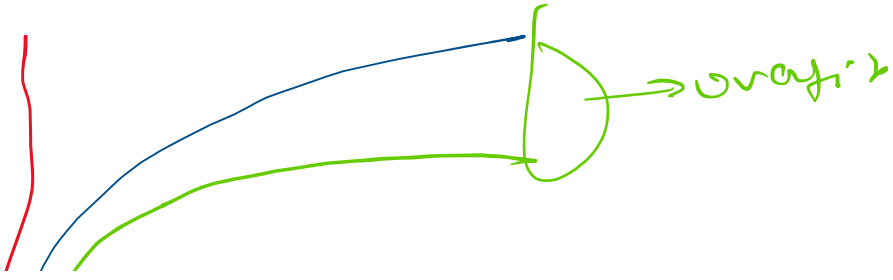
Simpler one →

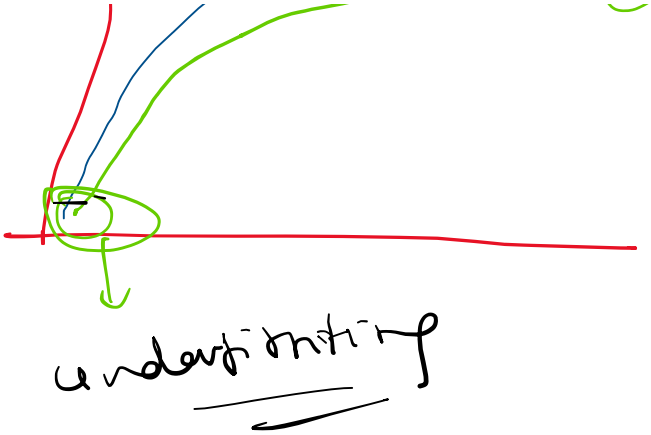
M2

→ less complex

→ pick simple things but not too simple







underfitting

Why is Occam's Razor important in machine learning?

21 users have participated

A	It helps in selecting the model that fits the training data perfectly.	28%
B	It encourages the use of complex models.	5%
C ✓	It helps in avoiding overfitting by favoring simpler models.	57%
D	It promotes the use of large datasets for training models.	0%

End Quiz Now

Does the model LR = 5*f1* + 0f1^2 + 0*f1*^2 + 0f1^4 underfit, overfit ?

4 users have participated

A ✓	Underfit	75%
B	Overfit	25%

End Quiz Now